

Introduction

The presence of a magnetic field splits the spin energy states. In this experiment, we determine the energy difference between these states and using this the magnetic moment μ_B of the electrons in the sample

Experiment Setup and Procedure

Experiment Setup

The sample is placed inside the tank coil of an RF oscillator, which is placed inside Helmholtz Coils.

Another RF oscillator generates RF waves near the sample. The current from the sample, through a series of amplifications and phase shifts, feeds into the oscilloscope where the waveform is displayed.

Procedure

First, the frequency of the external oscillator is adjusted so that it matches the frequency of the inner oscillator. At this resonance, the amplitude seen in oscilloscope surges rapidly. This can be used to determine the frequency.

Then, the external RF oscillator is set to a value sufficiently distant from the measured value, and AC current is passed through the Helmholtz Coils. This leads to formation of peaks in the oscilloscope reading. The time difference in the peaks is noted. Using this information, the magnetic moment is calculated.

Formulae and Relations

Energy difference in the two energy states is $\Delta E = g\mu_B H$

We are given $g = 2.04$

The sample is presence of photons of energy $\hbar\nu$ (where ν is the frequency of the RF oscillator).

So, when $\Delta E = \hbar\nu$ the electrons absorb photons and resonance is observed.

AC current of RMS value I and frequency ω is supplied to the Helmholtz Coils.

$$H(t) = \left(\frac{4}{5}\right)^{\frac{3}{2}} \frac{\mu_0 n I(t)}{a}$$

where n = number of turns in each coil and a is the radius of the coils.

$$I(t) = \sqrt{2} I \sin(\omega t)$$

If the time difference between the two peaks is $2t_0$ then we can say that at $t = t_0$,

$$\Delta E = \hbar\nu$$

$$\therefore g\mu_B H(t_0) = \hbar\nu$$

Note that here $I(t) = \sqrt{2}I\sin(\omega t_0)$

since all the quantities here are known, μ_B can be evaluated.

Data and Analysis

For the Helmholtz Coil:

$$n = 500$$

$$a = 7.7 \text{ cm} = 0.077 \text{ m}$$

$$\therefore H(t) = 0.00583 \times I(t)$$

Now when the frequency of the RF oscillator was measured to be

$$\nu = 13.75 \text{ MHz}$$

$$\hbar\nu = 9.11 \times 10^{-27} \text{ J}$$

The following reading for the time difference in the peaks observed was noted.

$I \text{ (A)}$	$2t_0 \text{ (ms)}$	$I(t_0) \text{ (A)}$	$H(t_0) \text{ (T)}$	$\mu_B \text{ (Am}^2\text{)}$
0.136	3.2	0.092	5.41×10^{-4}	8.25×10^{-24}
0.150	2.8	0.090	5.27×10^{-4}	8.46×10^{-24}
0.171	2.4	0.089	5.19×10^{-4}	8.59×10^{-24}
0.200	2.1	0.091	5.34×10^{-4}	8.34×10^{-24}

Now for a different frequency, the following data was noted.

$$\nu = 13.75 \text{ MHz}$$

$$\hbar\nu = 9.11 \times 10^{-27} \text{ J}$$

$I \text{ (A)}$	$2t_0 \text{ (ms)}$	$I(t_0) \text{ (A)}$	$H(t_0) \text{ (T)}$	$\mu_B \text{ (Am}^2\text{)}$
0.117	3.8	0.093	5.43×10^{-4}	8.67×10^{-24}
0.137	3.3	0.095	5.60×10^{-4}	8.40×10^{-24}
0.167	2.6	0.093	5.47×10^{-4}	8.59×10^{-24}
0.198	2.2	0.094	5.53×10^{-4}	8.50×10^{-24}